function, such as mean-squared error, to minimize the difference between the predicted value from our algorithm and the target prediction of $Q(S_t,A_t) + \alpha*[R_{t+1} + \gamma*\max Q(S_{t+1},A) - Q(S_t,A_t)]$.

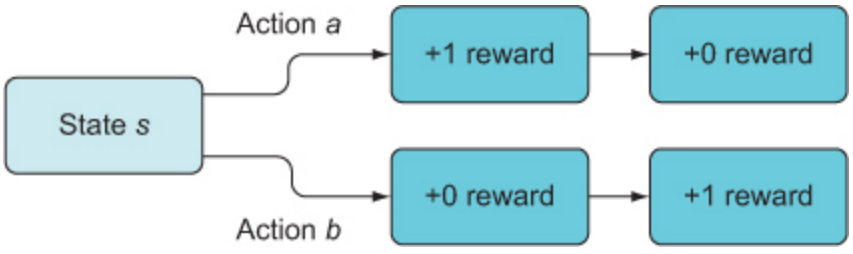
3.2.3. Hyperparameters

The parameters γ and α are called *hyperparameters* because they're parameters that influence how the algorithm learns but they're not involved in the actual learning. The parameter α is the *learning rate* and it's the same hyperparameter used to train many machine-learning algorithms. It controls how quickly we want the algorithm to learn from each move: a small value means it will only make small updates at each step, whereas a large value means the algorithm will potentially make large updates.

3.2.4. Discount factor

The parameter γ, the *discount factor*, is a variable between 0 and 1 that controls how much our agent discounts future rewards when making a decision. Let's take a simple example. Our agent has a decision between picking an action that leads to 0 reward then +1 reward, or an action that leads to +1 and then 0 reward (see figure 3.4).

Figure 3.4. An illustration of action trajectories leading to the same total reward but that may be valued differently since more recent rewards are generally valued more than distant rewards.



Previously, we defined the value of a trajectory as the expected reward. Both trajectories in figure 3.4 provide +1 overall reward though, so which sequence of actions should the algorithm prefer? How can we break the tie? Well, if the discount factor, γ, is less than 1, we will discount future rewards more than immediate rewards. In this simple case, even though both paths